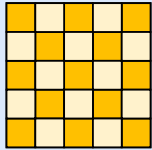
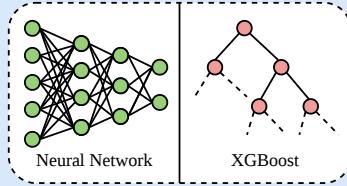


A

Train the baseline models on X_{train} , freeze them and evaluate on X_{test}


 X_{test}

evaluate



Neural Network

XGBoost

Baseline models

1

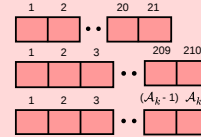
B

Action space over 21 retained APs

$$\mathcal{A}_k = \binom{N_{AP}}{k} = \binom{21}{k}$$

N_{AP} : total candidate APs (21)

k : APs chosen per action (attack size), $1 \leq k \leq N_{AP}$

 $\mathcal{A}_1 = 21$
 $\mathcal{A}_2 = 210$
 $\vdots$
 $\mathcal{A}_k$


2

C

Search strategies for subset identification

Brute Force
(ground truth)

Random Sampling

Simulated Annealing

RSSI-Proximity

Genetic Algorithm

Hill Climbing

DQN

3

E

Evaluation

$$\text{MSE}(\hat{\mathbf{y}}'_{\text{test}}, \mathbf{y}_{\text{test}}) = \frac{1}{N} \sum_{i=1}^N \|\hat{\mathbf{y}}'_{\text{test},i} - \mathbf{y}_{\text{test},i}\|^2$$

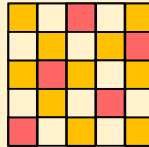
$\hat{\mathbf{y}}'_{\text{test}}$: predictions on attacked inputs X'_{test} ,

$\mathbf{y}_{\text{test}}$: ground truth

4

D

Apply noise η to the APs in selected subset S and evaluate the baseline model on the attacked data X'_{test}


 X'_{test}

$$X'_{\text{test}}(:, j) = \begin{cases} X_{\text{test}}(:, j) + \eta, & \text{if } j \in S, \\ X_{\text{test}}(:, j), & \text{if } j \notin S. \end{cases}$$